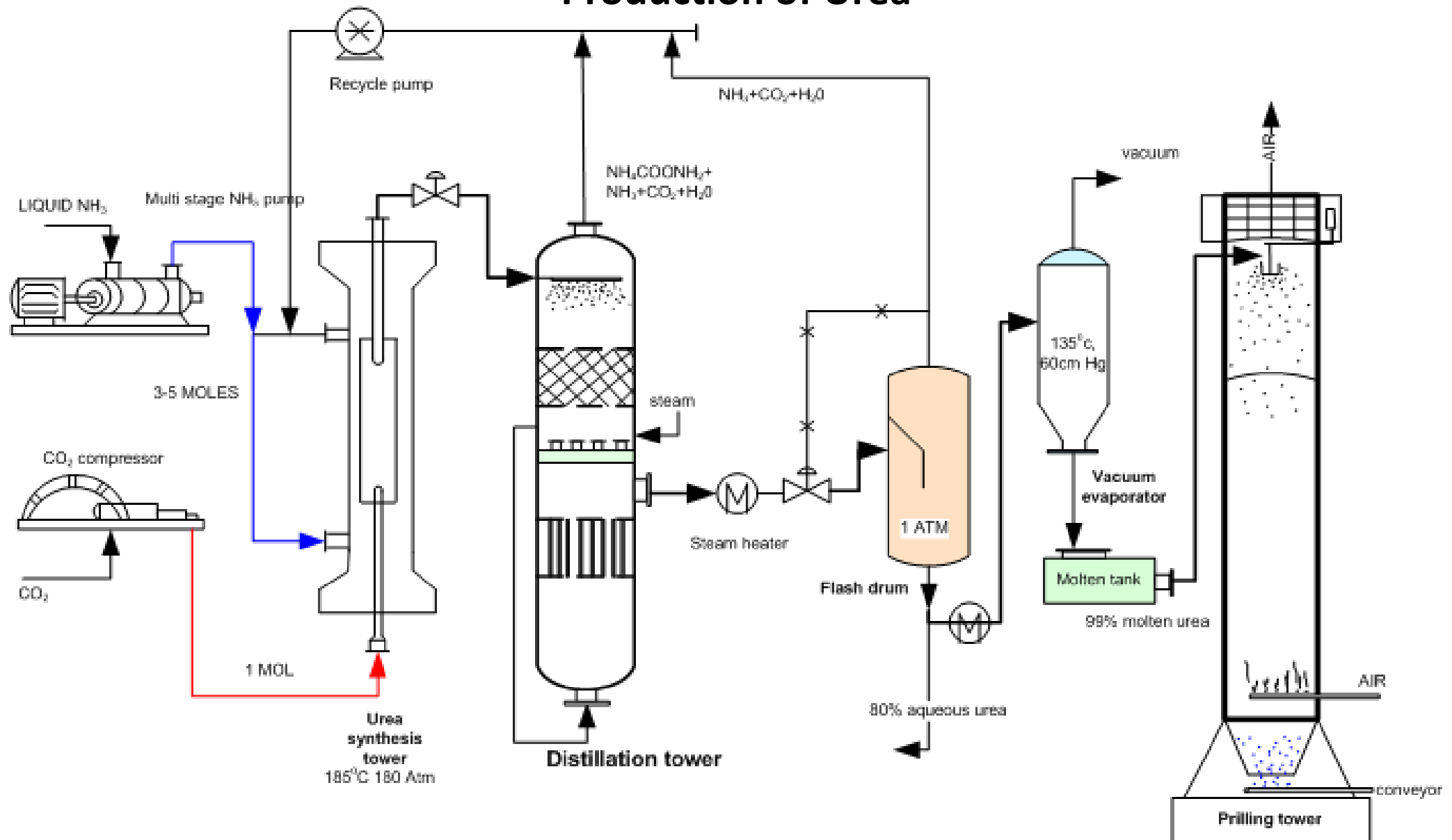


Computational Method in Chemical Engineering T2730C2

Textbooks:

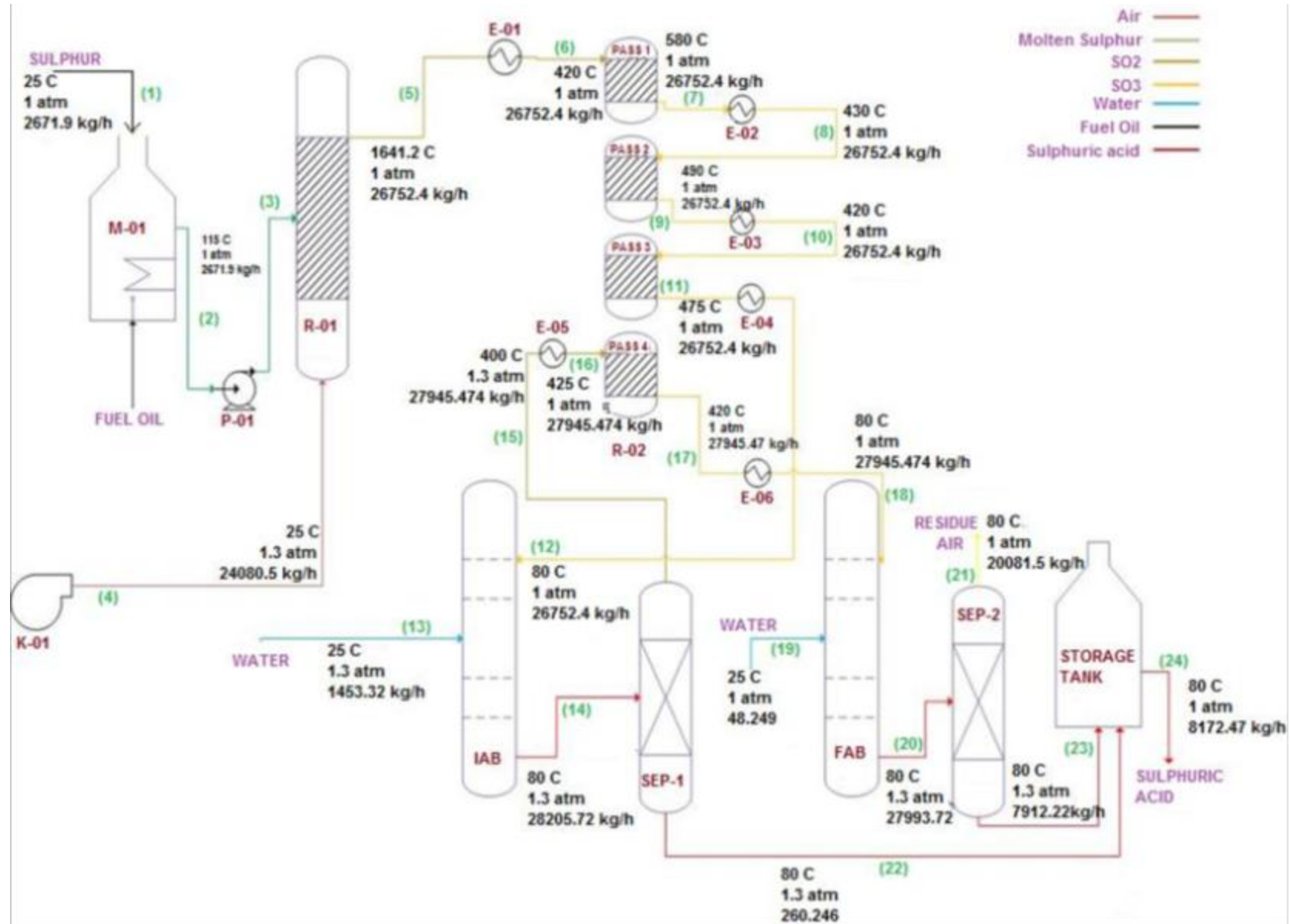
- Lorens T. Biegler, E. Ignacio Grossmann, Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International.
- Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis, and Evaluation, 2nd Edition, Wiley.
 - Robin Smith, Chemical Process: Design and Integration, Wiley.
- James M. Douglas, Conceptual Design of Chemical Processes, McGraw Hill International, 1988.

Production of Urea

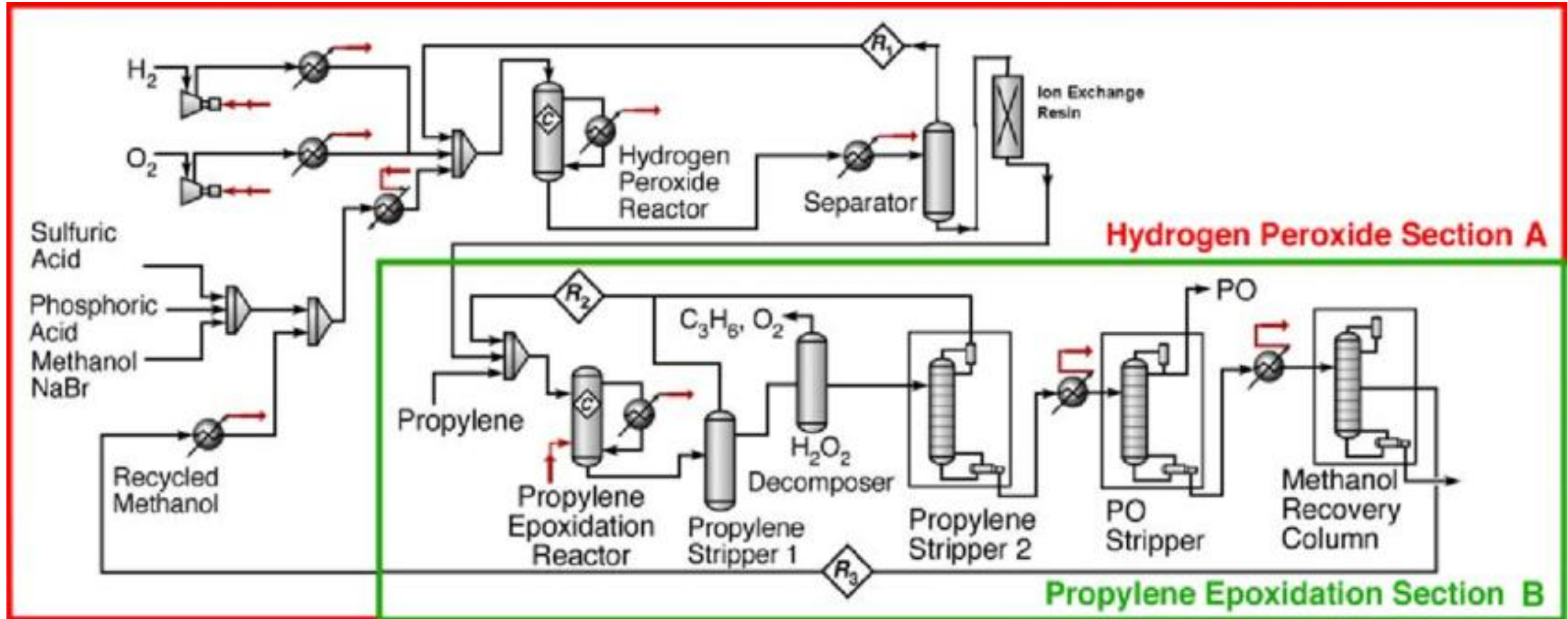


<https://enggyd.blogspot.com/2010/09/flow-diagram-of-urea-production-process.html>

Production of Sulfuric acid



Manufacture of Hydrogen Peroxide



https://www.researchgate.net/publication/303955897_Development_of_an_Economically_Viable_H2O2-based_Liquid-Phase_Ethylene_Oxide_Technology_Reactor_Engineering_and_Catalyst_Development_Studies

Unit 1 – The Design Process (Seider)

- Design Opportunities
- Steps in Product Process Design
- Environmental Protection
- Safety Considerations
- Engineering Ethics
- Role of Computers

Objectives of Design Process

- To know type of design decisions that challenge product and process design teams.
- To appreciate key steps in carrying out a product or process design.
- Be aware of environment issues and safety considerations prevalent in design of new chemical products and processes.
- Appreciate importance of maintaining high ethical principles in products and process design.
- Understand that chemical engineers use a blend of hand calculations, spreadsheets, mathematical computer packages, and process simulators to design products and processes.

Design Opportunities

- Design of chemical products begins with identification and creation of potential opportunities to satisfy societal needs and to generate profit.
- **Scope of chemical products:**
 1. **Basic chemical products** (speciality chemicals (ethylene, acetone, vinyl chloride, ethylene glycol, monomethyl ether), biomaterials (pharmaceuticals, tissue implants), polymeric materials (ethylene copolymers, polyvinyl chloride, polystyrene))
 2. **Industrial products** (films, fibers, papers)
 3. **Consumer products** (dialysis devices, ink-jet cartridges, fuel cells, cosmetics, detergents, pharmaceuticals)

Design Opportunities

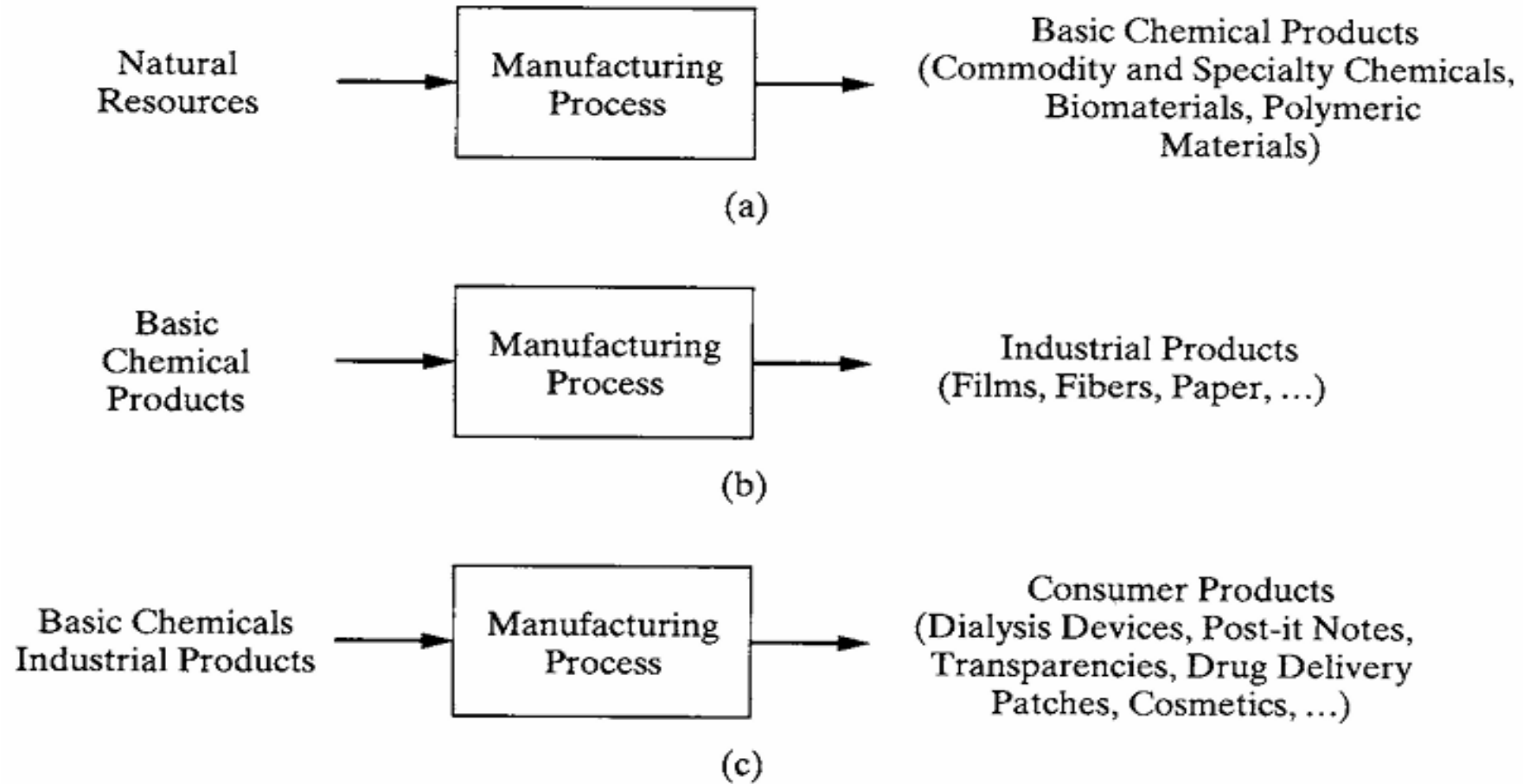


Figure 1.1 Manufacture of chemical products.

Design Opportunities - Basic chemical products

- Many chemical products are manufactured in small quantities and the design of a product focuses on identifying the chemicals or mixture of chemicals that have the desired properties, such as strength, stickiness, porosity, and permeability, to satisfy specific consumer needs.
- **For these, the challenge is to create a product that can be protected by patents and has sufficiently high market demand to command an attractive selling price.**
- **After the chemical mixture is identified, it is often necessary to design a manufacturing process.**

Design Opportunities - Industrial products

- Other chemical products, often referred to as commodity chemicals, **are required in large quantities.**
- These are often intermediates in the manufacture of specialty chemicals and industrial and consumer products.
- These include ethylene, propylene, butadiene, methanol, ethanol, ethylene oxide, ethylene glycol, ammonia, nylon, and caprolactam (for carpets), together with solvents like benzene, toluene, phenol, methyl chloride, and tetrahydrofuran, and fuels like gasoline, kerosene, and diesel fuel.
- **These are manufactured in large-scale processes that produce billions of pounds annually in continuous operation.**
- **Since they usually involve small well-defined molecules, the focus of the design is on the process to produce these chemicals from various raw materials.**

Steps in Product Process Design

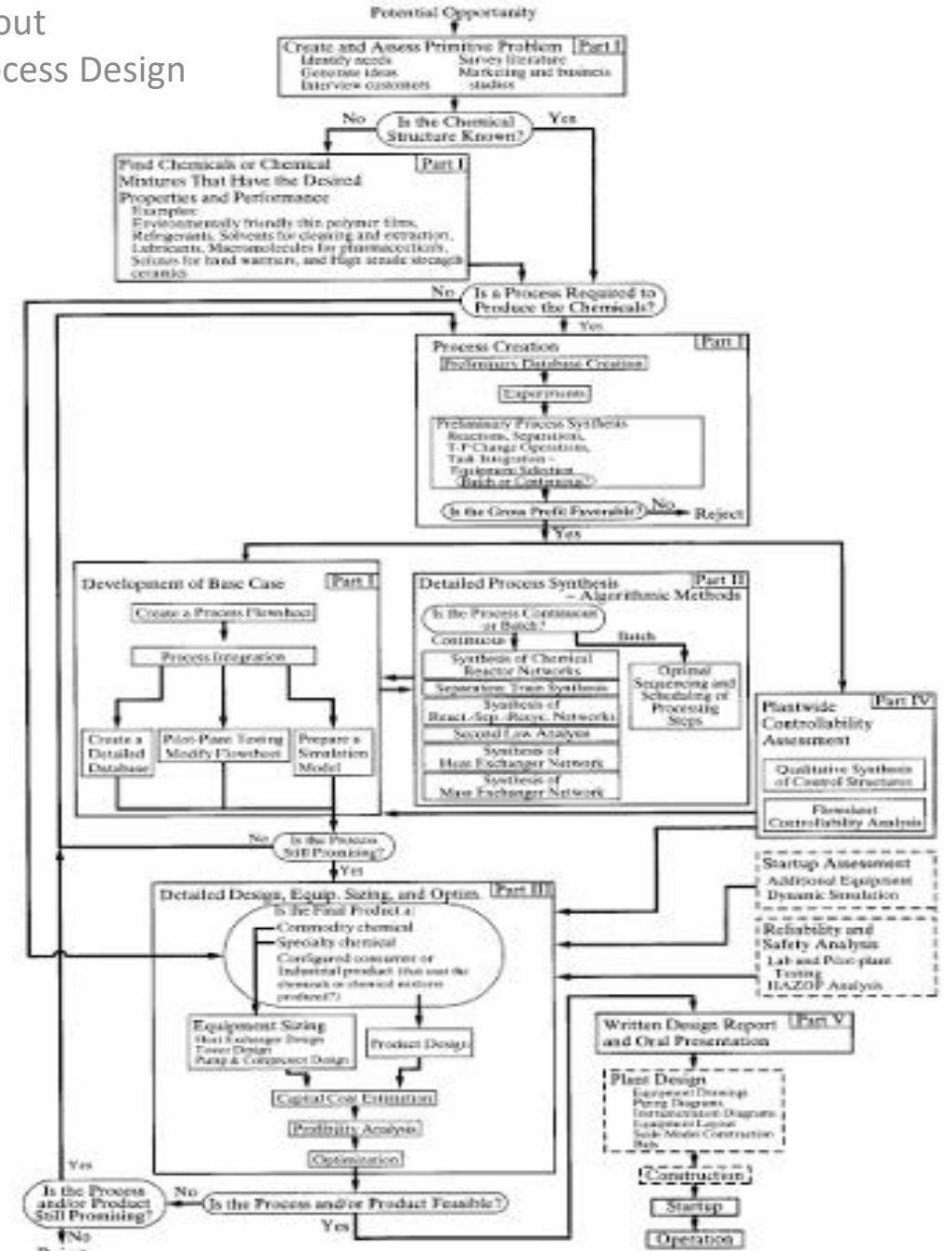
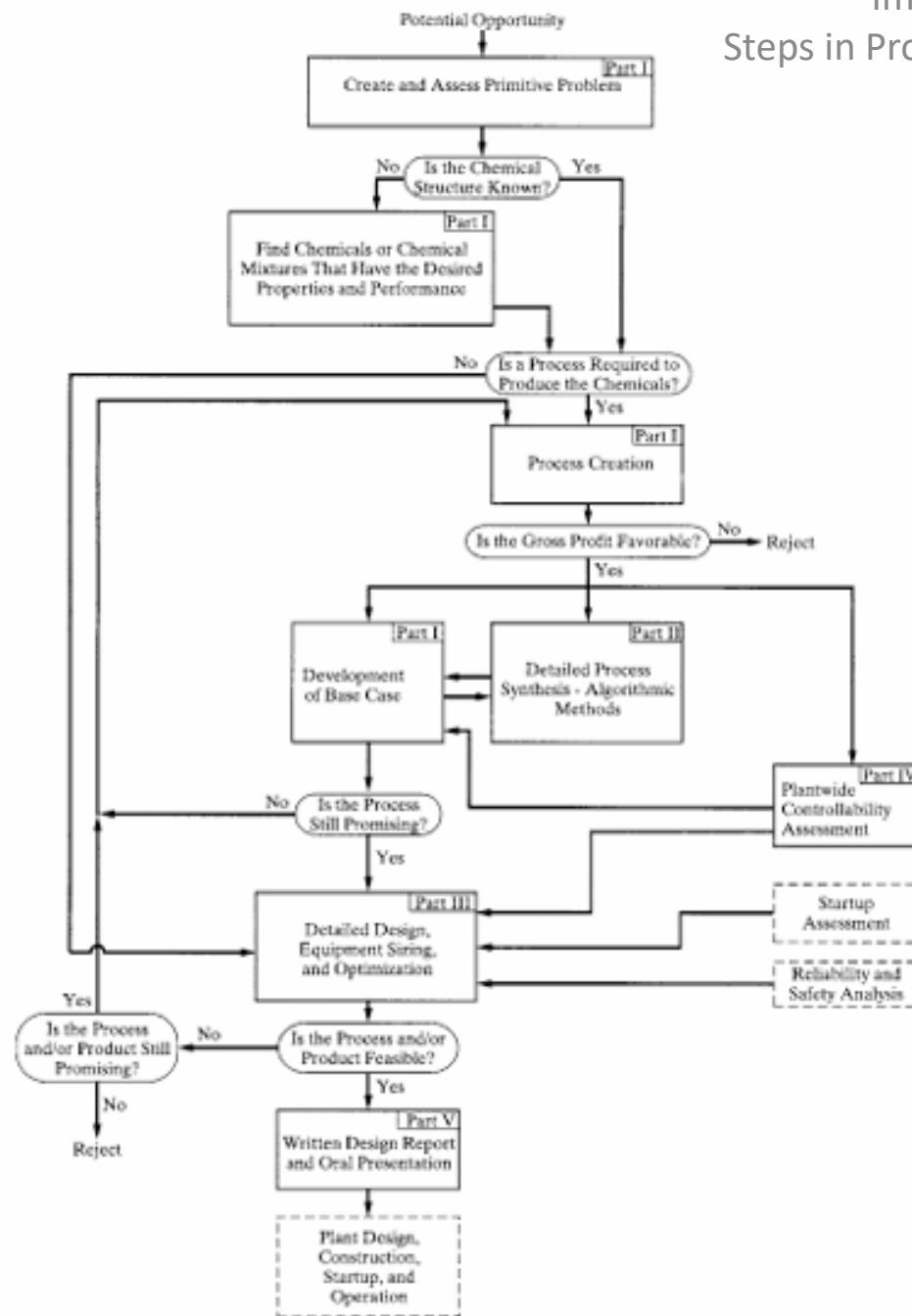
- Design problems are open ended and may have many solutions that are attractive and near optimal.
- No two designers design a complex product or process following exactly the same steps.

Challenges:

1. determining the composition of chemical mixtures to provide desired properties
 2. creating complex flowsheets and selecting operating conditions to produce desired products with a high degree of yield and selectivity, little recycle, and low utility costs
 3. creating configured industrial and consumer products.
- The creation of product and process designs are rarely straightforward and routine; rather they involve innovative approaches that lead to more profitable products and processes that are environmentally sound, and operationally safe.

Images in handout

Steps in Product and Process Design



Steps in Product Process Design

1. Create and assess primitive problem
2. Find chemicals or chemical mixtures having desired properties and performance
3. Process creation
4. Development of base case
5. Detailed process synthesis using algorithmic methods
6. Plantwide controllability assessment
7. Detailed design, equipment sizing, and optimization – configured product design
8. Written design report and oral presentation
9. Plant design, construction, startup, and operation

Create and assess primitive problem

References for ideas to begin engineering a product or process:

- Literature Survey
- SRI (consortium) Design Reports
- Encyclopedias (*Kirk-Othmer Encyclopedia of Chemical Technology* (1991) *Encyclopedia of Chemical Processing and Design* (McKetta and Cunningham, 1976), and *Ullmann's Encyclopedia of Industrial Chemistry* (1988), *McGraw-Hill Encyclopedia of Science and Technology* (1987)).
- Handbooks and reference books (*Perry's Chemical Engineer's Handbook* (Perry and Green, 1997), *the CRC Handbook of Chemistry and Physics*, *JANAF Thermochemical Tables* (Chase, 1985), *Riegel's Handbook of Industrial Chemistry* (Kent, 1992), *the Chemical Processing Handbook* (McKetta, 1993a), *the Unit Operations Handbook* (McKetta, 1993b), *Process Design and Engineering Practice* (Woods, 1995a), *Data for Process Design and Engineering Practice* (Woods, 1995b), *the Handbook of Reactive Chemical Hazards* (Bretherick, 1990), and *the Standard Handbook of Hazardous Waste Treatment and Disposal* (Freeman, 1989))

Create and assess primitive problem

References for ideas to begin engineering a product or process:

- Indexes – journals, technical reports, and books (*Applied Science and Technology Index, Engineering Index, Chemical Abstracts, Science Citation Index*)
- Patents
- Auxiliary Studies
- Tech Forum
- Process Innovation Technology Centers

Process Creation

- This involves the assembly of a **preliminary database** that is comprised of thermophysical property data, including vapor-liquid equilibrium data, flammability data, toxicity data, chemical prices, and related information needed for preliminary process synthesis.
- In some cases, experiments are needed to obtain important missing data that cannot be accurately estimated.
- Experimental reaction data are always required, as is experimental separation data when mixtures to be separated are moderately to highly nonideal.
- Then, preliminary process synthesis begins with the design team creating flowsheets involving just the reaction, separation, and temperature- and pressure-change operations.
- Process equipment is selected in task-integration step, which involves the selection of the operating mode - continuous, batch, or semicontinuous.
- Only those flowsheets that show a favorable gross profit are explored further.
- Detailed work on the process is avoided when the projected cost of the raw materials exceeds that of the products.

Development of a Base Case Process

- Development of just one flow diagram for each favorable process.
- **The design team begins by creating a detailed process flow diagram, accompanied by material and energy balances, and a list of the major equipment items.**
- A **material balance table** shows the state of each stream; the temperature, pressure, phase, flow rate, and composition, plus other properties.
- The material and energy balances are performed, at least in part, by a computer aided process simulator, such as **ASPEN PLUS, HYSYS, CHEMCAD, and PRO/II** for commodity chemicals, and **BATCH PLUS and SUPERPRO DESIGNER** for specialty chemicals, pharmaceuticals.
- Design team seeks opportunities to improve the designs of the process units and to achieve more efficient **process integrations** for the production of commodity chemicals, applying the methods of **heat and power integration**, for example, by exchanging heat between hot and cold streams, and **mass integration** to minimize raw materials and wastes.

Development of a Base Case Process

- Given the detailed process flow diagram, the design team refines the preliminary database to include additional data such as **transport properties and reaction kinetics, feasibilities of the separations, matches to be avoided in heat exchange (forbidden matches), heuristic parameters, equipment sizes and costs as a function of throughput.**
- This is usually accompanied by **pilot-plant testing** to confirm that the various equipment items will operate properly and to refine the database.
- If unanticipated data are obtained, the design team may need to revise the flow diagram.
- In some cases, equipment vendors run tests, as well as generate detailed equipment specifications.
- To complement these activities, a **simulation model** is prepared for the base-case design.
- *Process simulators are often useful in generating data bases because of their extensive data banks of pure-component properties and physical property correlations for ideal and nonideal mixtures.*
- When not available, simulation programs can regress experimental data taken in the laboratory or pilot plant for empirical or theoretical curve fitting.

Table 1.1 Process Operations for Three Types of Chemical Products

Product Type	Process Operation					
	1 Raw Materials Handling	2 Conversion	3	4	5 Quality Control	6 Packaging
Basic chemical products	Feeding, pumping, compressing	Chemical reaction	Purification: Separation	—	Intrinsic quality: Physical, thermal, chemical, rheological, properties	Shipped tank cars, pipelines
Industrial products	Feeding, pumping, web handling, drying, dehumidifying	Extrusion, blending, compounding	Primary forming: Die/profile extrusion, pultrusion, molding	Secondary forming: Sealing, stamping, thermal/light treatment, orientation	Functional quality: Optical weatherability, mechanical strength, printability	Packaged cartons, sacks, rolls
Consumer products	Web/fiber handling, belt conveying, discrete handling	Parts making includes die cutting, molding	Parts assembly and integration	Finishing	Functional (see above) and use quality: durability, lifetime, ease of use, effectiveness	Packaged individual containers, plastic wrapped, vials, bottles, cans

Environmental Protection

- **Environment Protection Agency (EPA)** in US to prevent air pollution, water pollution, prevent waste dumps containing toxic chemicals.
- Large investments have to be made by chemical process industries to eliminate sources of pollution.
- This increases cost of production which increases cost of end products.

Environmental Protection

Environmental issues:

1. Burning of fossil fuels for power generation and transportation (SO_2 , NO_x , CO, CO_2 , soot, ash, unburned hydrocarbons – causes acid rain (H_2SO_4), smog, hazes (NO_x), Greenhouse warming (CO_2), ozone layer reduction (NO_x , CO, formaldehyde, phenol).
2. Handling of toxic wastes – small portion (~3 %) is incinerated, bulk is disposed in land or in water (wastewater streams).
3. Bioaccumulated chemicals (insecticides DDT ($\text{C}_{14}\text{H}_9\text{Cl}_5$), solvent PCBs (polychlorinated biphenyls)).
4. Toxic metals and minerals (lead, mercury, cadmium, asbestos)

Environmental factors in process design

1. Reaction pathways to reduce byproduct toxicity (reference – Pocket Guide to Chemical Hazards of the National Institute for Occupational Safety and Health (NIOSH, 1987))
2. Reducing and reusing wastes (lifecycle of product and byproducts)
3. Avoiding nonroutine events (emphasis on steady state with reliable controllers and effective fault detection sensors, reduce transient operations, clean-up periods, catalysts regeneration cycles)
4. Materials characterization (need effective and rapid methods to measure concentration of toxic chemicals in gaseous or liquid effluent)
5. Design objectives, constraints, and optimization (formulate mixed objective functions to express environmental improvements in financial terms)
6. Regulations (EPA, EU, zoning regulations)

Environmental factors in process design

7. Intangible costs (avoid legal fees, public relations losses, complications related to public demonstrations)
8. Properties of dilute streams (ppb amounts, byproducts can react)
9. Properties of electrolytes (dissociate into ionic species – acid, bases, salts in dilute concentrations)

Safety Considerations

Safety issues

1. Fire and explosions

- LFL is the lower flammability limit (volume percent of the species in air below which flammability does not occur)
- UFL is the upper flammability limit (above which flammability does not occur).
- Within these limits, flames and explosions can occur and, consequently, design teams must be careful to keep the concentrations outside the flammability range.

Table 1.3 pertains to pure chemicals. For mixtures, the flammability limits are often estimated using the Le Chatelier equation, an empirical equation that must be used with caution:

$$\text{LFL}_{\text{mix}} = \frac{1}{\sum_{i=1}^C (y_i/\text{LFL}_i)} \quad \text{UFL}_{\text{mix}} = \frac{1}{\sum_{i=1}^C (y_i/\text{UFL}_i)} \quad (1.1)$$

Safety Considerations

Safety issues

1. Fire and explosions

- Autoignition temperatures: above which a flammable mixture is capable of extracting enough energy from the environment to self-ignite.
- At lower temperatures, an ignition source must be present.
- The flash point is the lowest temperature at which sufficient vapor exists in air to form an ignitable mixture.
- At the flash point, the vapor burns, but only briefly as insufficient vapor is formed to sustain combustion.

Safety Considerations

Safety issues

2. Toxic releases and dispersion models

- Design teams must identify ways in which toxic release can occur e.g. buildup of pressure in explosion, rupture of pipeline, collision of truck
- Select protective devices and processing units
- Model spread of dense and toxic vapor under different conditions to analyze chance of exposing surrounding community to serious health hazard

Design approaches towards safe chemical plants

1. Techniques to prevent fires and explosions (reduce oxygen concentration below minimum oxygen concentration (MOC), installation of grounding devices, explosion proof equipment and instruments, ventilation, sprinkler systems).
2. Relief devices (to relieve pressure, vented to atmosphere, vented to containment systems)
3. Hazards identification and risk assessment
4. Material safety data sheets (MSDS) (updated under OSHA (Occupational Safety and Health Agency))

Engineering Ethics

- Individual codes of ethics adopted by the various engineering societies (e.g., AIChE, ASCE, ASME, and IEEE) and by the NSPE.
- Code of Ethics adopted by the American Institute of Chemical Engineers (AIChE):
- Members of the American Institute of Chemical Engineers shall uphold and advance the integrity, honor, and dignity of the engineering profession by: being honest and impartial and serving with fidelity their employers, their clients, and the public; striving to increase the competence and prestige of the engineering profession; and using their knowledge and skill for the enhancement of human welfare. To achieve these goals, members shall:
 1. Hold paramount the safety, health, and welfare of the public in performance of their professional duties.
 2. Formally advise their employers or clients (and consider further disclosure, if warranted) if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public.

Engineering Ethics

3. Accept responsibility for their actions and recognize the contributions of others; seek critical review of their work and offer objective criticism of the work of others.
4. Issue statements and present information only in an objective and truthful manner.
5. Act in professional matters for each employer or client as faithful agents or trustees, and avoid conflicts of interest.
6. Treat fairly all colleagues and co-workers, recognizing their unique contributions and capabilities.
7. Perform professional services only in areas of their competence.
8. Build their professional reputations on the merits of their services.
9. Continue their professional development throughout their careers, and provide opportunities for the professional development of those under their supervision.

Role of Computers

- Many calculations in process design do not require detailed algorithms because they involve simple equations and graphic procedures that can be carried out quickly without the complications of computers.
- **Designers can make quick and effective decisions using heuristics and back-of-the-envelope calculations**
- In the earliest steps of process design, calculations are often quite approximate and the sources of data not very extensive.
- However, it does not take long for design teams to seek some computational assistance, particularly with process simulators, specially to access their extensive physical property data banks.
- As more data are obtained and flowsheets become more complicated, **designers use a combination of computer resources involving spreadsheets, mathematical packages, and process simulators, both steady-state and dynamic.**

Role of Computers

- Traditional procedural programming languages: **FORTRAN, C, C+ +, Java and Visual Basic.**
- Many colleges now require students to learn the use of such mathematical packages as **spreadsheets, MATLAB, POLYMATH, MATHCAD, and MAPLE; together with process simulators such as ASPEN PLUS, CHEMCAD, and HYSYS.Plant.**

Spreadsheets:

- Spreadsheets such as Excel, Lotus 1-2-3, and Quattro Pro have become as easy to use as word processors and communication packages.
- Most engineers can enter tables of data and program the spreadsheets to evaluate arithmetic expressions with very little effort.
- Whole columns and rows are manipulated and graphed, and results are stored and annotated without the complex formatting instructions that accompany procedural languages, even for the implementation of iterative algorithms.

Role of Computers

- Spreadsheets have added solvers for the solution of problems in nonlinear equations and optimization.
- In the design area, spreadsheets are widely used for profitability analysis, spreadsheets compute the total capital investment, the cost sheet, and various profitability measures, such as the return on investment and the net present value.
- Results from spreadsheets are readily plotted in any of a variety of graphs, and for the more difficult computational problems, a spreadsheet can be linked to a procedural language.
- Visual Basic for Applications (VBA) is available as a macro language for Microsoft Excel.

Role of Computers

Mathematical Packages

- Many kinds of engineering calculations can be carried out quickly and efficiently with symbolic and numeric mathematical packages.
- Examples are the analysis of linear systems (MATLAB), symbolic manipulations in algebra and calculus (MATHEMATICA, MAPLE), numerical integration of ordinary and partial differential equations (FEMLAB and MATLAB), and solution of mathematical programs in optimization (GAMS and LINDO).
- Optimization capabilities include the solution of linear programs (LPs), nonlinear programs (NLPs), mixed-integer linear programs (MILPs), and mixed-integer nonlinear programs (MINLPs).
- Using FORTRAN main programs, public domain software is available for the numerical solution of ordinary and partial differential equations (COLMOD, COLNEW, ODEPACK, and PDECOL) and for bifurcation analysis of nonlinear systems (AUTO).

Role of Computers

Process Simulators

- Computer-aided process design programs, often referred to as process simulators, flowsheet simulators, or flowsheeting packages, are widely used in process design.
- The facilities of the major simulators are more comprehensive than required for just **material and energy balances of a flowsheet with recycle and design specifications**.
- These packages are comprised of **data banks, physical property models, and equipment operation and sizing models**.
- The extensive data banks contain data on the **thermophysical and transport property** constants for thousands of chemicals.
- The simulators contain many **models of reactors and unit operations**, so-called simulation models that can be used to calculate material and energy balances.

Role of Computers

- Other models **compute equipment sizes and costs, and profitability measures.**
- Six major process simulators are widely used in the chemical process industries are **ASPEN PLUS, BATCH PLUS and HYSYS.Plant (Aspen Technology, Inc.), PRO/II (Simulation Sciences, Inc.), CHEMCAD (ChemStations, Inc.), and SUPERPRO DESIGNER (Intelligen, Inc.).**

Notes created by
Panchami Patel

Table 1.4 Computer Programs Useful in Process Design

1. Spreadsheets (commercial programs):
 - Corel Quattro Pro 10 in Word Perfect Office 2002 (<http://www.corel.com>)
 - Lotus 1-2-3 in Smart Suite 9.8 (<http://www.lotus.com>)
 - Microsoft Excel XP (<http://www.microsoft.com>)
2. Symbolic mathematics (commercial programs):
 - DERIVE 5 (<http://www.derive.com>)
 - MACSYMA (<http://www.mathtools.net/MATLAB/Symbolic/Other/>)
 - MAPLE 8 (<http://www.maplesoft.com>)
 - MATHCAD 11 (<http://www.mathcad.com>)
 - MATHEMATICA 4.2 (<http://www.wolfram.com>)
3. Numerical mathematics (commercial programs):
 - MACSYMA 2.4 (<http://www.scientek.com/macsyma/mxmain.htm>)
 - MAPLE 8 (<http://www.maplesoft.com>)
 - MATHCAD 11 (<http://www.mathcad.com>)
 - MATHEMATICA 4.2 (<http://www.wolfram.com>)
 - MATLAB 6.5 (<http://www.mathworks.com>)
 - POLYMATH 5.1 (<http://www.polymath-software.com>)
 - TK SOLVER 4 (<http://www.uts.com>)
4. Ordinary and Partial Differential Equations by numerical methods (public domain software that can be downloaded from the Internet site at <http://netlib2.cs.utk.edu>):
 - COLMOD An automatic continuation code for solving stiff boundary value problems.
 - COLNEW A general purpose code for solving mixed-order systems of boundary value problems using spline collocation and Newton's method.
 - ODEPACK A collection of codes for solving stiff and nonstiff systems of initial value problems.
 - PDECOL A code for solving coupled systems of partial differential equations of at most second order, in one space and one time dimension, or two space dimensions, using collocation on finite elements.

5. Ordinary and Partial Differential Equations by numerical methods (commercial programs):
FEMLAB 2.3 (<http://www.femlab.com>)
MATLAB 6.5 (<http://www.mathworks.com>)
6. Optimization (commercial programs):
LINDO APT 2.0 (<http://www.lindo.com>)
Microsoft Excel 2002 (<http://www.microsoft.com>)
GAMS (<http://www.gams.com>)
7. Bifurcation analysis (public domain program):
AUTO97 (<http://indy.cs.concordia.ca/auto>) A code for tracking by continuation the solution of systems of nonlinear algebraic and/or first-order ordinary differential equations as a function of a bifurcation parameter (available only for UNIX-based computers)
8. Process simulation, synthesis of distillation trains and heat-exchanger networks
Aspen Engineering Suite 11.1 (includes ASPEN PLUS, ASPEN DYNAMICS, ASPEN PINCH, ASPEN SPLIT, BATCH PLUS, etc.) and HYSYS.Plant
Aspen Technology, Inc., Ten Canal Park, Cambridge, MA 06141
Phone: 617-577-0100
FAX: 617-577-0303
<http://www.aspentec.com>
CHEMCAD 5.2.1
Chemstations, 2901 Wilcrest Drive, Suite 305, Houston, TX 77042
Phone: 713-978-7700
FAX: 713-978-7727
<http://www.chemstations.net>
Process Engineering Suite (includes PRO/II, HEXTRAN, etc.)
Simulation Sciences, Inc., 601 South Valencia Ave., Suite 100 Brea, CA 92823
Phone: 714-579-0412
FAX: 714-579-7927
<http://www.simsci.com>

SUPERPRO DESIGNER 4.9 (includes BatchPro, EnviroPro, and BioPro Designer)

Intelligen, Inc., 2326 Morse Avenue, Scotch Plains, NJ 07076

Phone: 908-654-0088

FAX: 908-654-3866

<http://www.intelligen.com/SuperPro.htm>

9. Stand-alone distillation packages

CHEMSEP 4.3 (<http://www.chemsep.org>)

Calculates continuous, multicomponent, multistage distillation, absorption, and stripping columns by equilibrium-stage and rate-based methods

MULTIBATCHDS 2.0 (<http://www.che.utexas.edu/cache/multibatch.html>)

Calculates multicomponent batch distillation

Quick Recap

- List scope of chemical products.
- What are two challenges in product process design?
- List any three steps in product process design.
- List some environment factors.
- List three design approaches for safe chemical plants.
- List two mathematical packages and two process simulators used to design chemical plants.